ASSIGNMENT-3

Real-time usage of supervised and unsupervised learning

Supervised Learning - Learns from past labeled examples to make predictions.

1. **Face Unlock in Smartphones**
   • Used by Apple (Face ID)
   • The system is trained with labeled face images.
   • It learns to recognize a specific person and unlocks only for them.
2. **Loan Approval System**
   • Used by banks like HDFC Bank
   • Trained on past data (approved/rejected loans).
   • Predicts whether a new loan application should be approved.
3. **Crop Disease Detection**
   • Used by agri-tech companies like CropIn
   • Images of crops are labeled (healthy/diseased).
   • The model predicts disease type from new images.

Unsupervised Learning - Discovers hidden patterns without predefined answers.

1. **News Topic Clustering**
   • Used by Google (Google News)
   • Groups similar news articles automatically.
   • No labels are given; the model finds patterns in content.
2. **Fraud Pattern Detection**
   • Used by Mastercard
   • Detects unusual spending patterns without predefined labels.
   • Flags transactions that differ from normal behavior.
3. **Customer Lifestyle Segmentation**
   • Used by Nike
   • Groups customers based on buying habits.
   • Helps target marketing campaigns effectively.

Open Router Parameters

- **Temperature** → Controls creativity (low = safe, high = creative)
- **Top P** → Chooses from most probable words (controls diversity)
- **Top K** → Limits selection to top K words
- **Min P** → Removes very low-probability words
- **Top A** → Advanced sampling control (rarely used)
- **Frequency Penalty** → Reduces repeated words
- **Presence Penalty** → Encourages new topics
- **Repetition Penalty** → Strong control to avoid repeating
- **Max Tokens** → Max total output length
- **Max Completion Tokens** → Max generated response length only
- **Logprobs** → Shows confidence of each word
- **Top Logprobs** → Shows alternative word choices
- **Seed** → Same input → same output (reproducible)
- **Logit Bias** → Force or block specific words